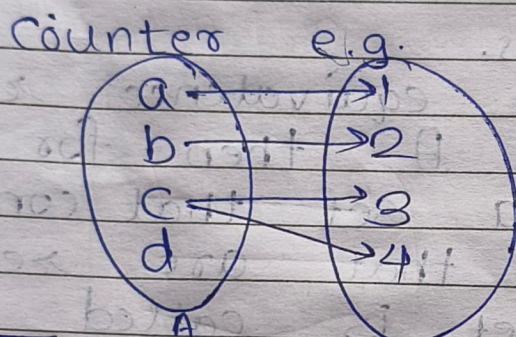
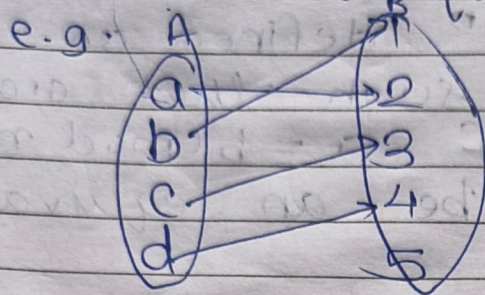


* Functions

A function f from a set A to set B , denoted by $f: A \rightarrow B$, is a rule that assigns to every element of A a unique element of B .



The above 'rel' is not a function for following reasons

- i) The element d of A is not related to any element of B .
- ii) The element c of A is assigned two elements of B .

* Types of functions

- i) One-one / Injective functions.

A function is said injective if every element of range of a function have a unique pre image.

Note: If $f: A \rightarrow B$ such that $a \in A$ is assigned $b \in B$ by f , then

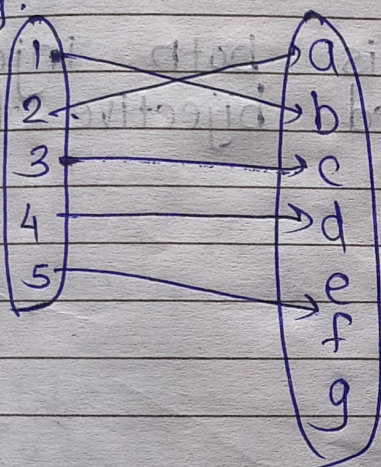
a) a is called the pre-image of b .
 b) b is called the image of a under f .
 And the above is represented by $f(a) = b$.

c) $A \rightarrow$ Domain of f

d) $B \rightarrow$ Co-domain of f

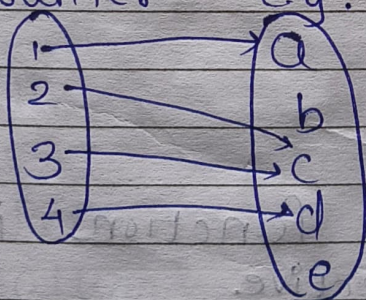
e) Set of all elements of B which have pre image is called the range of f .
 one-one.

e.g.



Counter

e.g.

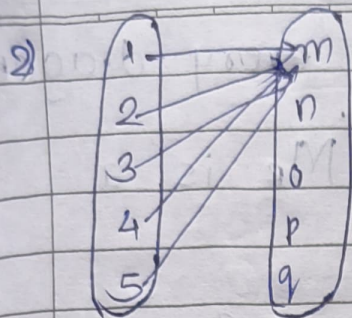


2) Onto / Surjective Functions:

A function f is said to be surjective if codomain of function is same as range of function.

$\exists$ = there exist

$\forall$ = such that



Range = $\{m\}$ $\neq$ codomain

$\therefore$ Not surjective

$m \xrightarrow{2} 2$
 $\downarrow$
 $1 \rightarrow 3$
 $\therefore$ Not injective

$\therefore$ Not bijective

3) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x$

$\rightarrow$ Consider $x \in \mathbb{R}$ we can observe that for any such ~~number~~ x , $\exists$ a unique $\frac{x}{2} \in \mathbb{R}$ such that $f(\frac{x}{2}) = 2 \times \frac{x}{2} = x$. Hence $\frac{x}{2}$ it is

4) $f: \mathbb{N} \rightarrow \mathbb{N}$ clear that any real no. x $f(x) = 5, \forall x \in \mathbb{N}$ has a pre image under f & that pre-image is unique. $\therefore f$ is both injective & surjective implying f is bijective.

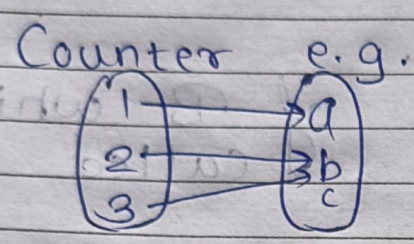
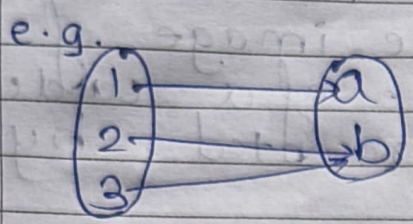
4) $f: \mathbb{N} \rightarrow \mathbb{N}$
 $f(x) = 5, \forall x \in \mathbb{N}$

$\rightarrow$ Function is neither injective nor surjective because;

- Range = $\{5\}$ which is not equal to codomain $\mathbb{N}$.
 - Pre image of 5 includes all natural numbers.
- $\therefore$ Function is not bijective.

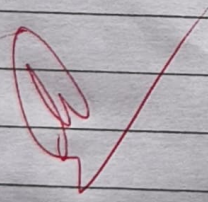
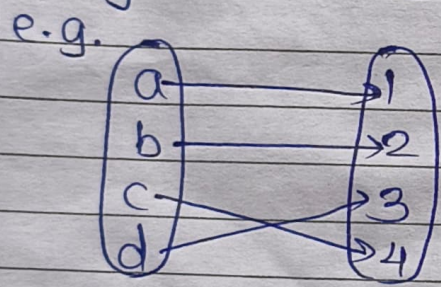
* Note:

Let R be a relⁿ defined on a set A and M_R denote the matrix representing the relⁿ R .

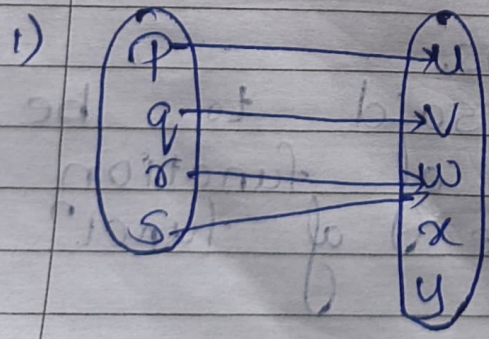


3) Bijective Functions

A function which is both injective & surjective is called bijective function.



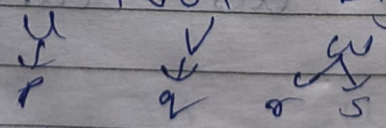
Q Check if following functions injective, surjective or bijective.



Range = {u, v, w} $\neq$ codomain.

Not surjective.

Not bijective.



$\therefore$ Not injective.

i) R is reflexive iff every diagonal entry of M_R is 1.

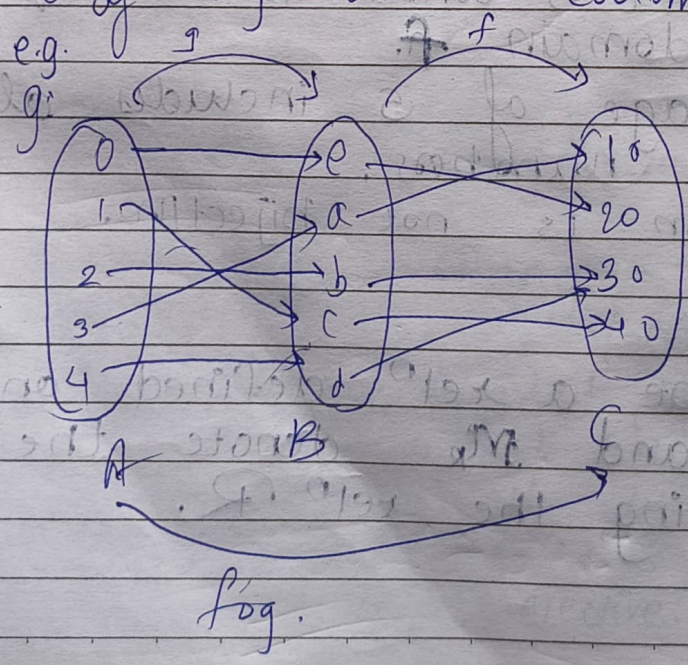
ii) R is symmetric iff M_R is a symmetric matrix.

* Composition of Relation & Function.

A composition of 2 functions f & g is an operation where 2 functions generate a new function denoted by $f \circ g$ in such a way that $f \circ g(x) = f(g(x))$ where g is a function from A to B , f is a function from B to C .

Note:-

- i) To define $f \circ g$ it is necessary that codomain of g is same as domain of f .
- ii) Whenever $f \circ g$ can be defined, domain of $f \circ g$ will be domain of g & codomain of $f \circ g$ will be codomain of f .



$$f \circ g(0) = f(g(0)) = f(e) = 20$$

$$f \circ g(1) = f(g(1)) = f(c) = 40$$

$\therefore$ codomain of $g =$ domain of f
 we can define $f \circ g$ as follows.

$$f \circ g : A \rightarrow C \text{ where}$$

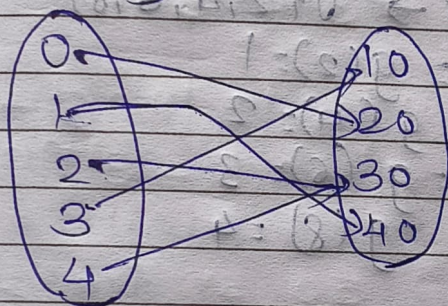
$$f \circ g(0) = f(g(0)) = f(e) = 20$$

$$f \circ g(1) = f(g(1)) = f(c) = 40$$

$$f \circ g(2) = f(g(2)) = f(b) = 30$$

$$f \circ g(3) = f(g(3)) = f(a) = 10$$

$$f \circ g(4) = f(g(4)) = f(d) = 30$$



Q. For following questions find $f \circ g$ & $g \circ f$, whenever they exist. In all cases of their existence specify domain & codomain of composition functions.

1) $f : \{1, 2, 3, 4\} \rightarrow \{2, 4, 6, 8\}$

$g : \{2, 4, 6, 8\} \rightarrow \{1, 2, 3, 4\}$

$f(1) = 2, f(2) = 4, f(3) = 6, f(4) = 8$

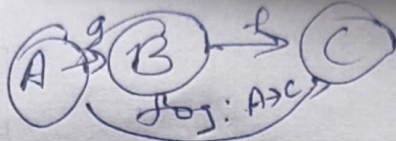
$g(2) = 1, g(4) = 2, g(6) = 3, g(8) = 4$

2) $f : \mathbb{R} \rightarrow \mathbb{R}$

$g : \mathbb{R} \rightarrow \mathbb{R}$

where $f(x) = 2x^2 - 3$

$g(x) = x^3$



$$(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$$

$$(a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

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YOUVA

3) $g: \mathbb{N} \rightarrow \mathbb{Z}$

$f: \mathbb{Z} \rightarrow \mathbb{R}$

$g(x) = 5x + 1$

$f(x) = x^3$

→ $f \circ g$: codomain of $f \rightarrow$ domain of f
 $\{2, 4, 6, 8\} \rightarrow \{1, 2, 3, 4\}$

$f \circ g(2) = f(g(2)) = f(1) = 1$

$f \circ g(4) = f(g(4)) = f(2) = 8$

$f \circ g(6) = f(g(6)) = f(3) = 27$

$f \circ g(8) = f(g(8)) = f(4) = 64$

$g \circ f$: codomain of $g \rightarrow$ domain of f
 $\{1, 2, 3, 4\} \rightarrow \{2, 4, 6, 8\}$

$g \circ f(1) = g(f(1)) = g(1) = 6$

$g \circ f(2) = g(f(2)) = g(8) = 41$

$g \circ f(3) = g(f(3)) = g(27) = 136$

$g \circ f(4) = g(f(4)) = g(64) = 325$

2) $f(x) = 2x^2 - 3$

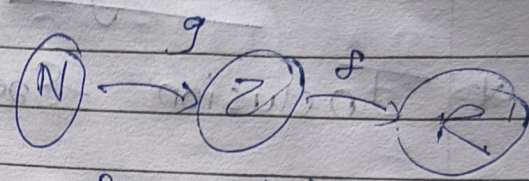
$g(x) = x^3$

→ $f \circ g(x)$
 $= f(g(x))$
 $= f(x^3)$
 $= 2(x^3)^2 - 3$
 $= 2x^6 - 6$

$g \circ f(x)$
 $= g(f(x))$
 $= g(2x^2 - 3)$
 $= (2x^2 - 3)^3$
 $= 8x^6 - 36x^4 + 54x^2 - 27$

3) $g(x) = 5x + 1, N \rightarrow Z$
 $f(x) = x^3, Z \rightarrow R$

$\rightarrow fog(x) = f(g(x))$



$\therefore f(5x+1)$
 $\therefore (5x+1)^3$
 $= 125x^3 + 1 + 75x^2 + 15x$

$g \circ f$ cannot be defined since codomain of f is not equal to domain of g .

But $f \circ g$ is defined & it is a function from $N \rightarrow R$

* Posets

Poset is a set with a relⁿ i.e. reflexive, anti-symmetric & transitive
 e.g. set of all +ve integers with divisibility relⁿ & set of all integers with " $\leq$ " relⁿ.